

Lista para entrega:

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Lista de Exercício 5:

02) $B = \{e_1, e_2, e_3\}$ e $B' = \{u, v, w\}$, sendo $(4, 0, 1)_B = (1, 2, 1)_{B'}$

$$B' \rightarrow B = \begin{bmatrix} -1 & 1 & x \\ 2 & -1 & y \\ 2 & 0 & z \end{bmatrix}, \text{ onde } x, y, z = ?$$

$$[v]_B = P_B + B' \cdot [v]_{B'},$$

logo:

$$\begin{bmatrix} -1 & 1 & x \\ 2 & -1 & y \\ 2 & 0 & z \end{bmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} (-1)(1) + 1(2) + x(1) \\ 2(1) + (-1)(2) + y(1) \\ 2(1) + 0(2) + z(1) \end{pmatrix}$$

$$\therefore \begin{pmatrix} x+1 \\ y \\ z+2 \end{pmatrix} = [v]_B \rightarrow \begin{cases} x+1 = 1 \rightarrow x=0 \\ y = 0 \rightarrow y=0 \\ z+2 = 1 \rightarrow z=-1 \end{cases}$$

Portanto, temos que: $x=0; y=0; z=-1$

04) Temos $B = (a, b, c)$ e $B' = (u, v, w)$.

$$\left. \begin{aligned} u &= (2, 1, 0)_B \rightarrow (2a + b + 0 \cdot c); \text{ logo } u = 2a + b \\ v &= (1, 1, 2)_B \rightarrow (a + b + 2c); \text{ " } v = a + b + 2c \\ w &= (0, 3, -2)_B \rightarrow (0 \cdot a + b - 2c); \text{ " } w = b - 2c \end{aligned} \right\} \text{ matriz } S$$

$$S = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 2 & -2 \end{pmatrix} \quad [x]_B = S[x]_{B'} \Rightarrow [x]_{B'} = S^{-1}[x]_B$$

$$S \cdot S^{-1} = I$$

$$\therefore \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 2 & -2 \end{bmatrix} \cdot \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} \bullet 2a + d &= 1 & \bullet 2b + e &= 0 & \bullet 2c + f &= 0 \\ \bullet a + d + g &= 0 & \bullet b + e + h &= 1 & \bullet c + f + i &= 0 \\ \bullet 2d + 2g &= 0 & \bullet 2e - 2h &= 0 & \bullet 2f - 2i &= 1 \end{aligned}$$

Logo, temos:

$$\begin{cases} 2a + d = 1 \\ a + d + g = 0 \\ 2d - 2g = 0 \end{cases}$$

$$\bullet d = \frac{1-d}{2}$$

$$\bullet \left(\frac{1-d}{2}\right) + \frac{d}{1} + \frac{g}{1} = 0$$

$$1-d+2d+2g=0$$

$$1+d+2g=0$$

$$g = \frac{-d-1}{2}$$

$$\bullet 2d - 2\left(\frac{-d-1}{2}\right) = 0$$

$$2d + d + 1 = 0$$

$$3d = -1$$

$$d = -\frac{1}{3}$$

$$a = \frac{1+\frac{1}{3}}{2} = \frac{\frac{4}{3}}{2} = \frac{2}{3}$$

$$g = \frac{\frac{1}{3}-1}{2} = \frac{-\frac{2}{3}}{2} = -\frac{1}{3}$$

$$\therefore S' = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{6} \\ -\frac{1}{3} & \frac{2}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{6} \end{pmatrix}$$

$$\begin{cases} 2b + e = 0 \\ b + e + h = 1 \\ 2e - 2h = 0 \end{cases}$$

$$\bullet b = \frac{-e}{2}$$

$$\bullet \frac{-e}{2} + e + h = 1$$

$$-e + 2e + 2h = 2$$

$$e + 2h = 2$$

$$2h = 2 - e$$

$$h = \frac{2-e}{2}$$

$$\bullet 2e - 2\left(\frac{2-e}{2}\right) = 0$$

$$2e - 2 + e = 0$$

$$3e = 2$$

$$e = \frac{2}{3}$$

$$b = \frac{-\frac{2}{3}}{2} = -\frac{2}{3} \cdot \frac{1}{2} = -\frac{1}{3}$$

$$h = \frac{2-\frac{2}{3}}{2} = \frac{\frac{4}{3}}{2} = \frac{2}{3}$$

$$\begin{cases} 2c + p = 0 \\ c + p + i = 0 \\ 2p - 2i = 1 \end{cases}$$

$$\bullet p = -2c$$

$$\bullet c - 2c + i = 0$$

$$-c + i = 0$$

$$i = c$$

$$\bullet 2(2c) - 2(c) = 1$$

$$-4c - 2c = 1$$

$$-6c = 1$$

$$c = -\frac{1}{6}$$

$$i = -\frac{1}{6}$$

$$p = -2\left(-\frac{1}{6}\right) = \frac{2}{6} = \frac{1}{3}$$

$$\therefore P_{B+B} = \begin{pmatrix} \frac{2}{3} & -\frac{1}{6} & -\frac{1}{6} \\ -\frac{1}{3} & \frac{2}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{6} \end{pmatrix}$$

10) $B_1 = \{b_1, b_2, b_3\} = \{(1, 0, 1), (1, 3, 2), (1, 2, 4)\}$, $v = (x, y, z)$

$B_2 = \{c_1, c_2, c_3\} = \{(1, 2, 1), (1, 3, 4), (2, 5, 6)\}$

$[B_1] = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & 2 & 4 \end{pmatrix}$

$[B_2] = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 3 & 5 \\ 1 & 4 & 6 \end{pmatrix}$

a) $[v]_{B_1} = \left(\begin{array}{ccc|c} 1 & 1 & 1 & x \\ 0 & 1 & 2 & y \\ 1 & 2 & 4 & z \end{array} \right) \xrightarrow{L_3 + L_3 - L_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & x \\ 0 & 1 & 2 & y \\ 0 & 1 & 3 & z-x \end{array} \right) \xrightarrow{L_3 + L_3 - L_2}$

$\leadsto \left(\begin{array}{ccc|c} 1 & 1 & 1 & x \\ 0 & 1 & 2 & y \\ 0 & 0 & 1 & z-x-y \end{array} \right) \therefore \begin{cases} c_3 = z-x-y \\ c_2 = y-2(z-x-y) = y+2x-2z \\ c_1 = x-(y+2x-2z)-(z-x-y) = -2y+z \end{cases}$

$[v]_{B_1} = \begin{bmatrix} -x-y+z \\ 2x+y-2z \\ -2y+z \end{bmatrix}$

$[v]_{B_2} = \left(\begin{array}{ccc|c} 1 & 1 & 2 & x \\ 2 & 3 & 5 & y \\ 1 & 4 & 6 & z \end{array} \right) \xrightarrow{L_2 + L_2 - 2L_1} \left(\begin{array}{ccc|c} 1 & 1 & 2 & x \\ 0 & 1 & 1 & y-2x \\ 0 & 3 & 4 & z-x \end{array} \right) \xrightarrow{L_3 + L_3 - 3L_2}$

$\leadsto \left(\begin{array}{ccc|c} 1 & 1 & 2 & x \\ 0 & 1 & 1 & y-2x \\ 0 & 0 & 1 & z-x-3(y-2x) \end{array} \right) \rightarrow z-x+3y+6x = 5x+3y+z$

$\therefore \begin{cases} c_3 = 5x+3y+z \\ c_2 = y-2x-(5x-3y+z) = -7x+4y-z \\ c_1 = x-2(5x-3y+z)-(-7x+4y-z) = -2x+2y-z \end{cases}$

$[v]_{B_2} = \begin{bmatrix} 5x+3y+z \\ -7x+4y-z \\ -2x+2y-z \end{bmatrix}$

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b) $\beta_1 \rightarrow \beta_2$

• $b_1 = (1, 0, 1)$ em β_2

$$\begin{pmatrix} 1 & 1 & 2 & | & 1 \\ 2 & 3 & 5 & | & 0 \\ 1 & 4 & 6 & | & 1 \end{pmatrix} \begin{array}{l} L_2 + L_2 - 2L_1 \\ L_3 + L_3 - L_1 \end{array} \rightsquigarrow \begin{pmatrix} 1 & 1 & 2 & | & 1 \\ 0 & 1 & 1 & | & -2 \\ 0 & 3 & 4 & | & 0 \end{pmatrix} \begin{array}{l} \\ L_3 + L_3 - 3L_2 \end{array}$$

$$\begin{pmatrix} 1 & 1 & 2 & | & 1 \\ 0 & 1 & 1 & | & -2 \\ 0 & 0 & 1 & | & 6 \end{pmatrix} \therefore \begin{cases} \alpha_1 = 1 - \alpha_2 - 2\alpha_3 = 1 - (-8) - 6 = 3 \\ \alpha_2 = -2 - \alpha_3 = -8 \\ \alpha_3 = 6 \end{cases}$$

Logo: $\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = \begin{pmatrix} 3 \\ -8 \\ 6 \end{pmatrix}$

• $b_2 = (1, 1, 2)$ em β_2

$$\begin{pmatrix} 1 & 1 & 2 & | & 1 \\ 2 & 3 & 5 & | & 1 \\ 1 & 4 & 6 & | & 2 \end{pmatrix} \begin{array}{l} L_2 + L_2 - 2L_1 \\ L_3 + L_3 - L_1 \end{array} \rightsquigarrow \begin{pmatrix} 1 & 1 & 2 & | & 1 \\ 0 & 1 & 1 & | & -1 \\ 0 & 3 & 4 & | & 1 \end{pmatrix} \begin{array}{l} \\ L_3 + L_3 - 3L_2 \end{array}$$

$$\begin{pmatrix} 1 & 1 & 2 & | & 1 \\ 0 & 1 & 1 & | & -1 \\ 0 & 0 & 1 & | & 4 \end{pmatrix} \therefore \begin{cases} \beta_1 = 1 - \beta_2 - 2\beta_3 = 1 - (-5) - 2(4) = -2 \\ \beta_2 = -1 - \beta_3 = -5 \\ \beta_3 = 4 \end{cases}$$

Logo: $\begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \begin{pmatrix} -2 \\ -5 \\ 4 \end{pmatrix}$

• $b_3 = (1, 2, 4)$ em β_2

$$\begin{pmatrix} 1 & 1 & 2 & | & 1 \\ 2 & 3 & 5 & | & 2 \\ 1 & 4 & 6 & | & 4 \end{pmatrix} \begin{array}{l} L_2 + L_2 - 2L_1 \\ L_3 + L_3 - L_1 \end{array} \rightsquigarrow \begin{pmatrix} 1 & 1 & 2 & | & 1 \\ 0 & 1 & 1 & | & 0 \\ 0 & 3 & 4 & | & 3 \end{pmatrix} \begin{array}{l} \\ L_3 + L_3 - 3L_2 \end{array}$$

$$\begin{pmatrix} 1 & 1 & 2 & | & 1 \\ 0 & 1 & 1 & | & 0 \\ 0 & 0 & 1 & | & 3 \end{pmatrix} \therefore \begin{cases} \gamma_1 = 1 - \gamma_2 - 2\gamma_3 = 1 - 6 = -2 \\ \gamma_2 = -\gamma_3 = -3 \\ \gamma_3 = 3 \end{cases}$$

Logo: $\begin{pmatrix} \gamma_1 \\ \gamma_2 \\ \gamma_3 \end{pmatrix} = \begin{pmatrix} -2 \\ -3 \\ 3 \end{pmatrix}$

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Portanto: $P_{B_1 \rightarrow B_2} = \begin{pmatrix} \alpha_1 & \beta_1 & \gamma_1 \\ \alpha_2 & \beta_2 & \gamma_2 \\ \alpha_3 & \beta_3 & \gamma_3 \end{pmatrix} = \begin{pmatrix} 3 & -2 & -2 \\ -8 & -5 & -3 \\ 6 & 4 & 3 \end{pmatrix}$

c) $P_{B_2 \rightarrow B_1}$ • $b'_1 = (1, 2, 1)$ em B_1

$$\begin{pmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 1 & 2 & | & 2 \\ 1 & 2 & 4 & | & 1 \end{pmatrix} \xrightarrow{L_3 + L_1 - L_2} \begin{pmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 1 & 2 & | & 2 \\ 0 & 1 & 3 & | & 0 \end{pmatrix} \xrightarrow{L_3 + L_2 - L_1}$$

$$\begin{pmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 1 & 2 & | & 2 \\ 0 & 0 & 1 & | & -2 \end{pmatrix} \therefore \begin{cases} \alpha_1 = 1 - \alpha_2 - \alpha_3 = -3 \\ \alpha_2 = 2 - 2(\alpha_3) = 2 + 4 = 6 \\ \alpha_3 = -2 \end{cases}$$

Logo: $\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = \begin{pmatrix} -3 \\ 6 \\ -2 \end{pmatrix}$

• $b'_2 = (1, 3, 4)$ em B_1

$$\begin{pmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 1 & 2 & | & 3 \\ 1 & 2 & 4 & | & 4 \end{pmatrix} \xrightarrow{L_3 + L_1 - L_2} \begin{pmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 1 & 2 & | & 3 \\ 0 & 1 & 3 & | & 3 \end{pmatrix} \xrightarrow{L_3 + L_2 - L_1}$$

$$\begin{pmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 1 & 2 & | & 3 \\ 0 & 0 & 1 & | & 0 \end{pmatrix} \therefore \begin{cases} \beta_1 = 1 - 3 - 0 = -2 \\ \beta_2 = 3 - 0 = 3 \\ \beta_3 = 0 \end{cases}$$

Logo: $\begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix}$

• $b'_3 = (2, 5, 6)$ em B_1

$$\begin{pmatrix} 1 & 1 & 1 & | & 2 \\ 0 & 1 & 2 & | & 5 \\ 1 & 2 & 4 & | & 6 \end{pmatrix} \xrightarrow{L_3 + L_1 - L_2} \begin{pmatrix} 1 & 1 & 1 & | & 2 \\ 0 & 1 & 2 & | & 5 \\ 0 & 1 & 3 & | & 4 \end{pmatrix} \xrightarrow{L_3 + L_2 - L_1}$$

$$\begin{pmatrix} 1 & 1 & 1 & | & 2 \\ 0 & 1 & 2 & | & 5 \\ 0 & 0 & 1 & | & -1 \end{pmatrix} \therefore \begin{cases} \gamma_1 = 2 - \gamma_3 - \gamma_2 = -4 \\ \gamma_2 = 5 - 2(\gamma_3) = 5 + 2 = 7 \\ \gamma_3 = -1 \end{cases}$$

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$$\text{Logo: } \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -4 \\ 7 \\ -1 \end{pmatrix}$$

$$\text{Portanto: } P_{B_2} \rightarrow P_{B_1} = \begin{pmatrix} \alpha_1 & \beta_1 & \gamma_1 \\ \alpha_2 & \beta_2 & \gamma_2 \\ \alpha_3 & \beta_3 & \gamma_3 \end{pmatrix} = \begin{pmatrix} -3 & -2 & -4 \\ 6 & 3 & 7 \\ -2 & 0 & -1 \end{pmatrix}$$

$$16) a) T: (x, y) = T(\alpha x_1, \alpha y_1)$$

$$= (\alpha x_1 + \alpha y_1, \alpha x_1 - \alpha y_1) \quad \text{Homogeneidade } \checkmark$$

$$= (\alpha(x_1 + y_1), \alpha(x_1 - y_1)) \quad \text{Linear } \checkmark$$

$$= \alpha(T(x, y))$$

$$b) T: \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$(x, y) \mapsto xy$$

$$T(u+v) = (x_1+x_2)(y_1+y_2) = x_1y_1 + x_1y_2 + x_2y_1 + x_2y_2$$

$$T(u) + T(v) = x_1y_1 + x_2y_2$$

Aditividade X

Logo não é linear.

$$c) T: M_2(\mathbb{R}) \rightarrow \mathbb{R}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \det \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Aditividade X

$$T(u+v) = T(x_1+y_1, x_2+y_2)$$

Não é linear

$$\det(A+B) \neq \det(A) + \det(B)$$

$$d) T: P_2(\mathbb{R}) \rightarrow P_3(\mathbb{R})$$

$$ax^2 + bx + c \mapsto ax^3 + bx^2 + cx$$

$$T(u+v) = T(a_1+a_2)x^2 + (b_1+b_2)x + (c_1+c_2)$$

$$(a_1+a_2)x^3 + (b_1+b_2)x^2 + (c_1+c_2)x$$

$$= (a_1x^3 + b_1x^2 + c_1x) + (a_2x^3 + b_2x^2 + c_2x)$$

$$= T(u+v)$$

Aditividade $\checkmark$

$$\begin{aligned} T(du) &= T(d(ax^e + bx^e + c)) = (dax^e + dbx^e + dca) \\ &= d(ax^e + bx^e + ca) \\ &= dT(u) \end{aligned}$$

Homogeneidade ✓

Linear ✓

$$2) T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$$

$$(x, y) \mapsto (x-y, 3x, -2y)$$

$$\begin{aligned} T(u+v) &= T(x_1+x_2, y_1+y_2) \\ &= ((x_1+x_2) - (y_1+y_2), 3(x_1+x_2), -2(y_1+y_2)) \\ &= ((x_1-y_1), 3x_1, -2y_1) + ((x_2-y_2), 3x_2, -2y_2) \\ &= T(u) + T(v) \end{aligned}$$

Aditividade ✓

$$T(du) = T(dx_1, dy_1)$$

$$= (d(x_1+y_1), d(3x_1), d(-2y_1))$$

$$= (d((x_1+y_1)), 3x_1, -2y_1)$$

$$= d(T(u))$$

Homogeneidade ✓

∴ Linear ✓

$$3) T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$(x, y, z) \mapsto (x+y, x-y, 0)$$

$$\begin{aligned} T(u+v) &= T(x_1+x_2, y_1+y_2, z_1+z_2) \\ &= (x_1+y_1, x_1-y_1, 0) + (x_2+y_2, x_2-y_2, 0) \\ &= T(u) + T(v) \end{aligned}$$

Aditividade ✓

$$T(du) = T(dx_1, dy_1, dz_1)$$

$$= (d(x_1+y_1), d(x_1-y_1), d(0))$$

$$= d((x_1+y_1), (x_1-y_1), 0)$$

$$= d(T(u))$$

Homogeneidade ✓

∴ Linear ✓

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$$g) T: \mathbb{R}^2 \rightarrow \mathbb{R}^4$$

$$(x, y) \mapsto (y, x, y, x)$$

$$T(u+v) = T(x_1+x_2, y_1+y_2)$$

$$= (y_1+y_2, x_1+x_2, y_1+y_2, x_1+x_2)$$

$$= (y_1, x_1, y_1, x_1) + (y_2, x_2, y_2, x_2)$$

$$= T(u) + T(v)$$

Aditividad ✓

$$T(\alpha u) = T(\alpha x, \alpha y)$$

$$= (\alpha y, \alpha x, \alpha y, \alpha x)$$

$$= \alpha (y, x, y, x)$$

$$= \alpha (T(u))$$

Homogeneidad ✓

∴ Linear ✓

$$h) T: \mathbb{R}^2 \rightarrow M_2(\mathbb{R})$$

$$(x, y) \mapsto \begin{pmatrix} 2y & 3x \\ -y & x+2y \end{pmatrix}$$

$$T(u+v) = T(x_1+x_2, y_1+y_2)$$

$$= \begin{pmatrix} 2(y_1+y_2) & 3(x_1+x_2) \\ -(y_1+y_2) & (x_1+x_2)+2(y_1+y_2) \end{pmatrix}$$

$$= \begin{pmatrix} 2y_1 & 3x_1 \\ -y_1 & x_1+2y_1 \end{pmatrix} + \begin{pmatrix} 2y_2 & 3x_2 \\ -y_2 & x_2+2y_2 \end{pmatrix}$$

$$= T(u) + T(v)$$

Aditividad ✓

$$T(\alpha u) = T(\alpha x, \alpha y)$$

$$\begin{pmatrix} \alpha 2y_1 & \alpha 3x_1 \\ \alpha (-y_1) & \alpha (x_1+2y_1) \end{pmatrix} = \alpha \begin{pmatrix} 2y_1 & 3x_1 \\ -y_1 & x_1+2y_1 \end{pmatrix}$$

$$= \alpha (T(u))$$

Homogeneidad ✓

∴ Linear ✓

i) $T: M_2(\mathbb{R}) \rightarrow \mathbb{R}^2$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto (a-c, b+c)$$

Sejam $A = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}$ e $B = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}$ e $\alpha \in \mathbb{R}$

$$T(A+B) = T \begin{pmatrix} a_1+a_2 & b_1+b_2 \\ c_1+c_2 & d_1+d_2 \end{pmatrix}$$

$$= ((a_1+a_2)-(c_1+c_2), (b_1+b_2)+(c_1+c_2))$$

$$= (a_1+c_1, b_1+c_1) + (a_2+c_2, b_2+c_2)$$

$$= T(A) + T(B)$$

Aditividade ✓

$$T(\alpha u) = T \begin{pmatrix} \alpha a_1 & \alpha b_1 \\ \alpha c_1 & \alpha d_1 \end{pmatrix}$$

$$= (\alpha a_1 - \alpha c_1, \alpha b_1 + \alpha c_1)$$

$$= \alpha (a_1 - c_1, b_1 + c_1)$$

$$= \alpha (T(u))$$

Homogeneidade ✓

∴ Linear ✓

Lista 6

6a) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3, T(x, y, z) = (x+2y-z, y+z, x+y-2z)$

$$T(1, 0, 0) = (1, 0, 1)$$

$$T(0, 1, 0) = (2, 1, 1)$$

$$T(0, 0, 1) = (-1, 1, -2)$$

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 1 & 1 & -2 \end{pmatrix}$$

$$\rightsquigarrow \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{pmatrix}$$

$L_3 + L_2 - L_1$

$L_3 + L_2 + L_2$

$$\begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

combinações lineares

$$\text{Im}(A) = \{(1, 0, 1), (2, 1, 1)\}$$

Kerrel:

$$\begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{L_1 + L_1 - 2L_2} \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$A \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \quad \begin{cases} x - 3z = 0 \\ y + z = 0 \\ 0 = 0 \text{ sendo } z = t \end{cases} \sim \begin{cases} x = 3t \\ y = -t \\ z = t \end{cases}$$

$\therefore$ base por $\ker(t) = \{3, -1, 1\}$

b) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2, T(x, y) = (x+y, x+y)$

$$O = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, T(1, 0) = (1, 1)$$

$$T(0, 1) = (1, 1)$$

$\text{Im}g = \{(1, 1), (1, 1)\} \rightarrow \text{Im}g = \{(1, 1)\}$

Kerrel:

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{cases} x + y = 0 \rightarrow x = -(-t) = t \rightarrow t(1, -1) \\ x + y = 0 \rightarrow y = -t \end{cases}$$

$\therefore \ker(t) = \{(1, -1)\}$

14) $F: \mathbb{R}^4 \rightarrow \mathbb{R}^3$

$$F(x, y, z, t) = (x - y + z + t, x + 2z - t, x + y + 3z - 3t)$$

$$t(1, 0, 0, 0) = (1, 1, 1)$$

$$t(0, 1, 0, 0) = (-1, 0, 1)$$

$$t(0, 0, 1, 0) = (1, 2, 3)$$

$$t(0, 0, 0, 1) = (1, -1, -3)$$

$$A = \begin{pmatrix} 1 & -1 & 1 & 1 \\ 1 & 0 & 2 & -1 \\ 1 & 1 & 3 & -3 \end{pmatrix} \xrightarrow{\begin{matrix} L_2 \rightarrow L_2 - L_1 \\ L_3 \rightarrow L_3 - L_1 \end{matrix}} \begin{pmatrix} 1 & -1 & 1 & 1 \\ 0 & 1 & 1 & -2 \\ 0 & 2 & 2 & -4 \end{pmatrix} \xrightarrow{L_3 \rightarrow L_3 - 2L_2}$$

$$\begin{pmatrix} 1 & -1 & 1 & 1 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{matrix} \text{rank}(A) = 2 \\ \dim(\text{Im}g) = 2 \end{matrix}$$

$t(1,0,0) = (1,1,1)$, logo a base de $\text{Im}(F)$ é: $\{(1,1,1), (-1,0,1)\}$
 $t(0,1,0) = (-1,0,1)$

• Kernel:

$$\begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{cases} x + 2y + t = 0 \\ x + y - 2z = 0 \end{cases} \leadsto \begin{cases} x = -2y - t \quad (z=0) \\ y = -x + 2z \quad (t=0) \end{cases}$$

$\therefore (x, y, z, t) = s(-2, 1, 1, 0) + u(1, 2, 0, 1)$

base de $\text{Ker}(F) = \{(-2, 1, 1, 0), (1, 2, 0, 1)\}$

16) a) $A = \begin{pmatrix} -3 & 4 \\ 1 & 2 \end{pmatrix}$

$$X_A(\lambda) = \det(A - \lambda I) = \det \begin{pmatrix} -3 - \lambda & 4 \\ -1 & 2 - \lambda \end{pmatrix}$$

$$= (-3 - \lambda)(2 - \lambda) - 4(-1)$$

$$= -6 + 3\lambda + (-2\lambda) + \lambda^2 + 4$$

$$= \lambda^2 + \lambda - 2 \leadsto (\lambda - 1)(\lambda + 2)$$

$\therefore$ as autovalores são: $\lambda_1 = 1$ e $\lambda_2 = -2$

p/ $\lambda = 1$ $(A - I)V = 0$ $\begin{pmatrix} -3-1 & 4 \\ -1 & 2-1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -4 & 4 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$ $\begin{cases} -4x + 4y = 0 \\ -x + y = 0 \end{cases} \leadsto x = y$ logo, $v_1 = (1, 1)$	p/ $\lambda = -2$ $(A + 2I)V = 0$ $\begin{pmatrix} -3+2 & 4 \\ -1 & 2+2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 & 4 \\ -1 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$ $\begin{cases} -x + 4y = 0 \\ -x + 4y = 0 \end{cases}$ logo, $v_2 = (-1, 4)$
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b) $A = \begin{pmatrix} \sqrt{3} & -1 \\ 1 & \sqrt{3} \end{pmatrix}$

$$X_A(\lambda) = \det \begin{pmatrix} \sqrt{3} - \lambda & -1 \\ 1 & \sqrt{3} \end{pmatrix} = (\sqrt{3} - \lambda)^2 + 1$$

$$= \lambda^2 - 2\sqrt{3}\lambda + 3 + 1 = \lambda^2 - 2\sqrt{3}\lambda + 4 \leadsto \begin{cases} \lambda_1 = \sqrt{3} + i \\ \lambda_2 = \sqrt{3} - i \end{cases}$$

(2)

$\therefore$ autovalores: $\lambda_1 = \sqrt{3} - i$ e $\lambda_2 = \sqrt{3} + i$

$p/\lambda = \sqrt{3} - i$

$$A - \lambda I = \begin{pmatrix} 3 - (\sqrt{3} - i) & -1 \\ 1 & \sqrt{3} - (\sqrt{3} - i) \end{pmatrix} = \begin{pmatrix} i & -1 \\ 1 & i \end{pmatrix}$$

$p/\lambda = \sqrt{3} + i$

$$A - \lambda I = \begin{pmatrix} -i & -1 \\ 1 & -i \end{pmatrix}$$

$$\begin{cases} ix - y = 0 \rightarrow y = ix \rightarrow y = i \\ x + iy = 0 \rightarrow x = 1 \end{cases}$$

$$\therefore v_1 = \begin{pmatrix} 1 \\ i \end{pmatrix}$$

$$\begin{cases} -x - y = 0 \rightarrow y = -x \\ x - yi = 0 \rightarrow x = 1 \end{cases}$$

$$\therefore v_2 = \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

c) $A = \begin{pmatrix} 3 & 0 & -4 \\ 0 & 3 & 5 \\ 0 & 0 & -1 \end{pmatrix}$

Autovalores são elementos da diagonal

$$\lambda_1 = 3 \text{ e } \lambda_2 = -1$$

Triangular superior

$p/\lambda = -1$

$$A + (-1)I = A + I = \begin{pmatrix} 4 & 0 & -4 \\ 0 & 4 & 5 \\ 0 & 0 & 0 \end{pmatrix}$$

$p/\lambda = 3$

$$(A - 3I)u = 0$$

$$A - 3I = \begin{pmatrix} 3-3 & 0 & -4 \\ 0 & 3-3 & 5 \\ 0 & 0 & -1-3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & -4 \\ 0 & 0 & 5 \\ 0 & 0 & -4 \end{pmatrix}$$

$$\begin{cases} 4x - 4y = 0 \rightarrow x = y \\ 4y + 5z = 0 \rightarrow y = -\frac{5}{4}z \end{cases}$$

$$\text{logo } z = 4, x = 4 \text{ e } y = -5$$

$$\therefore v_1 = (4, -5, 4)$$

$$\begin{cases} -4z = 0 \\ -5z = 0 \\ -4z = 0 \end{cases} \quad z = 0$$

$$(x, y, 0) = x(1, 0, 0) + y(0, 1, 0)$$

$$\therefore v_2 = \{(1, 0, 0), (0, 1, 0)\}$$

d) $A = \begin{pmatrix} 3 & -3 & 4 \\ 0 & 3 & 5 \\ 0 & 0 & -1 \end{pmatrix}$

Autovalores:

$$\lambda_1 = 3 \text{ e } \lambda_2 = -1$$

Triangular superior

(13)

$$p/\lambda = 3$$

$$A - 3I = \begin{pmatrix} 3-3 & -3 & 4 \\ 0 & 3-3 & 5 \\ 0 & 0 & -1-3 \end{pmatrix} = \begin{pmatrix} 0 & -3 & 4 \\ 0 & 0 & 5 \\ 0 & 0 & -4 \end{pmatrix}$$

$$\begin{cases} -3y + 4z = 0 \\ 5z = 0 \rightarrow z = 0 \\ -4z = 0 \end{cases}$$

$$\therefore (x, 0, 0) = x(1, 0, 0)$$

$$v_1 = (1, 0, 0)$$

$$p/\lambda = 3$$

$$A + I = \begin{pmatrix} 4 & -3 & 4 \\ 0 & 4 & 5 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\begin{cases} 4x - 3y + 4z = 0 \\ 4y + 5z = 0 \rightarrow y = -\frac{5}{4}z \\ 0 = 0 \end{cases}$$

$$4x - 3\left(-\frac{5}{4}z\right) + 4z = 4x + \frac{15}{4}z + 4z =$$

$$= 4x + \frac{31}{4}z = 0 \rightarrow x = -\frac{31}{16}z$$

$$\therefore x = -31 \quad z = 20$$

$$v_2 = (31, 20, -16)$$

Lista 07:

$$1) f(v_1, v_2) = 2x_1y_1 + x_1y_2 + y_1x_2 + x_2y_2$$

• 1º) provar simetria:

$$f(v_1, v_2) = f(v_2, v_1), \text{ p/ qualquer } v_1 = (x_1, x_2); v_2 = (y_1, y_2)$$

$$f(v_1, v_2) = f(v_2, v_1)$$

$$2x_1y_1 + x_1y_2 + y_1x_2 + x_2y_2 = 2y_1x_1 + y_1x_2 + x_1y_2 + y_2x_2$$

Como, a multiplicação em $\mathbb{R}$ é comutativa,

$$f(v_1, v_2) = f(v_2, v_1), \text{ logo simetria } \checkmark$$

• 2º) provar bilinearidade

- Sejam $u = (u_1, u_2), v = (v_1, v_2), w = (w_1, w_2) \in \mathbb{R}^2$ e $\alpha, \beta \in \mathbb{R}$.

- Considerando $\alpha u + \beta v = (\alpha u_1 + \beta v_1, \alpha u_2 + \beta v_2)$

$$f(\alpha u + \beta v, w) = 2(\alpha u_1 + \beta v_1)w_1 + (\alpha u_1 + \beta v_1)w_2 + \dots$$

$$\dots (\alpha u_2 + \beta v_2)w_2 + (\alpha u_2 + \beta v_2)w_1$$

$$f(\alpha u + \beta v, w) = \alpha(2u_1w_1 + u_1w_2 + u_2w_1 + u_2w_2) + \dots$$

$$\dots \beta(2v_1w_1 + v_1w_2 + v_2w_1 + v_2w_2) + \dots$$

$$\dots \alpha f(u, w) + \beta f(v, w)$$

f é linear pelo 1º argumento e, pelo simetria, é linear no 2º argumento. Logo, f é bilinear $\checkmark$

3º positividade definida:

$$v = (x_1, x_2) \in \mathbb{R}^2$$

$$f(v, v) \geq 0$$

$$\begin{aligned} f(v, v) &= 2x_1x_1 + x_1x_2 + x_1x_2 + x_2x_2 \\ &= 2x_1^2 + 2x_1x_2 + x_2^2 \end{aligned}$$

$$2x_1^2 + 2x_1x_2 + x_2^2 = (x_1^2) + (x_1^2 + 2x_1x_2 + x_2^2)$$

$$x_1^2 \geq 0 \quad \text{e} \quad x_1^2 + 2x_1x_2 + x_2^2 = (x_1 + x_2)^2 \geq 0$$

logo,

$$f(v, v) = x^2 + (x_1 + x_2)^2 \geq 0$$

$$x_1^2 = 0 \rightarrow x_1 = 0$$

$$(x_1 + x_2)^2 = 0 \rightarrow x_1 + x_2 = 0,$$

como $x_1 = 0$ e $x_2 = 0$, logo $v = (0, 0)$

$$f(v, v) = 0, \text{ só p/ } v = 0,$$

portanto é positivo definido ✓

A função f satisfaz todas as propriedades, logo é um produto interno.

$$6) \langle u, v \rangle = x_1y_1 - 2x_1y_2 - 2x_2y_1 + 5x_2y_2$$

• 1º Simetria:

$$\langle u, v \rangle \rightarrow \langle v, u \rangle$$

$$\langle u, v \rangle = y_1x_1 - 2y_1x_2 - 2y_2x_1 + 5y_2x_2$$

Pela comutatividade em $\mathbb{R}$, é simétrico ✓

• 2º Bilinearidade:

Sejam $u = (x_1, x_2)$, $v = (x'_1, x'_2)$, $w = (y_1, y_2)$ e $\alpha, \beta \in \mathbb{R}$

$$\alpha u + \beta v = (\alpha x_1 + \beta x'_1, \alpha x_2 + \beta x'_2)$$

$$\langle \alpha u + \beta v, w \rangle = (\alpha x_1 + \beta x'_1)y_1 - 2(\alpha x_1 + \beta x'_1)y_2 - \dots$$

$$\dots - 2(\alpha x_2 + \beta x'_2)y_1 + 5(\alpha x_2 + \beta x'_2)y_2$$

$$= \alpha(x_1y_1 - 2x_1y_2 - 2x_2y_1 + 5x_2y_2) + \beta(x'_1y_1 - 2x'_1y_2 - 2x'_2y_1 + 5x'_2y_2)$$

$$= \alpha \langle u, w \rangle + \beta \langle v, w \rangle$$

bilinear ✓

3° positividade definida.

$$\begin{aligned}\langle u, u \rangle &= x_1^2 - 2x_1x_2 - 2x_2x_1 + 5x_2^2 + 5x_2^2 \\&= x_1^2 + 4x_1x_2 + 5x_2^2 \\&= x_1^2 - 4x_1x_2 + 5x_2^2 \\&= (x_1 - 2x_2)^2 + x_2^2\end{aligned}$$

logo:

$$(x_1 - 2x_2)^2 \geq 0 \text{ e } x_2^2 \geq 0, \text{ assim } \langle u, u \rangle \geq 0 \text{ p/4 } u$$

Portanto é produto interno em $\mathbb{R}^2$.

II) $u_1 = (1, 1, 1)$

$$\|u_1\| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$c_1 = \frac{\langle u_1, u_1 \rangle}{\|u_1\|} = \frac{1}{\sqrt{3}} \cdot (1, 1, 1)$$

• projetando u_2 em u_1

$$\langle u_2, u_1 \rangle = 1 \cdot 1 + (-1) \cdot (1) + 1 \cdot 1 = 1 - 1 + 1 = 1$$

$$\text{proj}_{u_1}(u_2) = \frac{\langle u_2, u_1 \rangle}{\langle u_1, u_1 \rangle} \cdot u_1 = \frac{1}{3} \cdot (1, 1, 1)$$

$$\begin{aligned}u_2 &= v_2 - \text{proj}_{u_1}(v_2) = (1, -1, 1) - \frac{1}{3}(1, 1, 1) \\&= (1 - \frac{1}{3}, -1 - \frac{1}{3}, 1 - \frac{1}{3}) \\&= \left(\frac{2}{3}, -\frac{4}{3}, \frac{2}{3}\right) \\&= \frac{2}{3}(1, -2, 1)\end{aligned}$$

$$\|u_2\| = \frac{2}{3} \cdot \sqrt{1^2 + (-2)^2 + 1^2} = \frac{2}{3} \cdot \sqrt{6} = \frac{2\sqrt{6}}{3}$$

$$c_2 = \frac{\langle u_2, u_2 \rangle}{\|u_2\|} = \frac{\frac{2}{3} \cdot (1, -2, 1)}{\frac{2\sqrt{6}}{3}} = \frac{1}{\sqrt{6}}(1, -2, 1)$$

(16)

• projetor u_3 em u_2

$$\begin{aligned}\langle u_3, u_2 \rangle &= \langle (-1, 0, 1), \frac{2}{3}(1, -2, 1) \rangle \\ &= \frac{2}{3}((-1) \cdot 1 + 0 \cdot (-2) + 1 \cdot 1) = \frac{2}{3} \cdot (-1 + 0 + 1) = 0\end{aligned}$$

$$\text{proj}_{u_2}(u_3) = 0$$

$$u_3 = u_3 - \text{proj}_{u_1}(u_3) - \text{proj}_{u_2}(u_3) = u_3 = (-1, 0, 1)$$

$$\|u_3\| = \sqrt{(-1)^2 + 0^2 + (1)^2} = \sqrt{2}$$

$$e_3 = \frac{u_3}{\|u_3\|} = \frac{1}{\sqrt{2}}(-1, 0, 1)$$

• , temos: $e_1 = \frac{1}{\sqrt{3}}(1, 1, 1)$

$$e_2 = \frac{1}{\sqrt{6}}(1, -2, 1)$$

$$e_3 = \frac{1}{\sqrt{2}}(-1, 0, 1)$$

